

Hack2Heal 2.0

Global Healthcare Hackathon



SENTINEL — Community Outbreak Early Warning System

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SENTINEL — Community Outbreak Early Warning System

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Core Solution: A zero-cost outbreak early-warning system combining icon-based reporting, vernacular voice capture (Vosk/Bhashini), and spatiotemporal cluster analysis to detect disease patterns in rural areas.

Addressing the Problem: Replaces complex, text-heavy tools with voice notes and messaging apps rural communities already use, connecting isolated household symptoms to alert local health centers before outbreaks spread.

Innovation & Uniqueness: Tracks multi-stage symptom *progression* instead of single keywords, and integrates automated WHO verbal autopsies to capture vital mortality data from unrecorded deaths.

Technologies Used

Intake & Mobile: Flutter (icon-based UI), Telegram Bot API (voice/text).

Speech & AI: Bhashini, Vosk, Google Gemini API.

Backend & Analytics: Python (FastAPI), spatiotemporal scan statistics.

Storage & Hosting: Supabase (PostgreSQL + PostGIS), Redis, Render/Vercel.

Methodology & Implementation

Voice/Icon Reporting: Captures local dialect voice notes or tap-inputs without requiring literacy.

AI Structuring: Transcribes vernacular audio and parses inputs into standardized symptom stages.

Spatiotemporal Engine: Scans geographic progression patterns to detect micro-clusters before hospitals fill up.

WHO Verbal Autopsy: Gathers voice-based illness histories for unrecorded deaths to recover lost mortality data.

Automated Escalation: Triggers real-time alerts to local Primary Health Centres and district dashboards.

FEASIBILITY AND VIABILITY



Feasibility

Adoption: Uses pre-existing habits (voice notes on messaging apps) with zero training curve.

Cost: Operates entirely on free-tier cloud tools and India's open Bhashini AI.

Challenges & Risks

Adoption: Low literacy levels and rural smartphone discomfort.

Data Quality: Regional dialect diversity and background audio noise.

Operations: False alarms overloading Primary Health Centre staff.

Cultural: Navigating bereavement sensitively during post-mortem reporting.

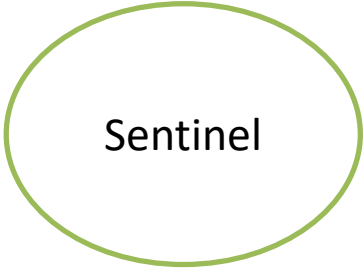
Mitigation Strategies

Zero-Text UX: Icon-first app paired with vernacular voice capture.

Hybrid STT: Vosk offline engine backed by Bhashini speech models.

Progression Filters: Alerts trigger only on multi-case symptom progressions, not raw counts.

WHO Framework: Low-burden, respectful voice flows designed specifically for grieving families.



Sentinel

IMPACT AND BENEFITS



Target Audience Impact

Villagers: Report illness instantly via voice in local dialects without needing literacy or digital skills.

ASHA Workers: Gain early cluster warnings, replacing guesswork with clear, targeted intervention areas.

Primary Health Centres: Receive real-time geographic alerts, enabling rapid containment before clinics overflow.

Key Benefits

Social: Protects vulnerable rural families and recovers critical cause-of-death data from unrecorded fatalities.

Economic: Cuts outbreak response expenses and operates on a zero-recurring-cost cloud infrastructure.

Systemic: Modernizes rural disease surveillance by linking remote villages directly to district health systems.

References & Research Work

WHO Verbal Autopsy Standards: Built on the *WHO Verbal Autopsy Instrument* (WHO Standards for Ascertaining and Attributing Causes of Death) for structured post-mortem voice capture. Link: [who.int](https://www.who.int)

Spatiotemporal Outbreak Detection: Implements principles from *Martin Kulldorff's Space-Time Scan Statistics* (SaTScan methodology) for localized epidemiological cluster detection. Link: [satscan.org](https://www.satscan.org)

National Disease Surveillance Framework: Aligned with India's *Integrated Disease Surveillance Programme (IDSP)* guidelines for rural district reporting. Link: idsp.mohfw.gov.in

Indic Speech AI Infrastructure: Powered by *Project Bhashini* (National Language Translation Mission, MeitY) for regional voice transcription. Link: bhashini.gov.in

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
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